

Tab 1

SIH CALCULATION & MODEL JUSTIFICATION REPORT

Closed-Container Dataset

1. Summary

The workbook implements a complete calculation. The measured signal starts as a 10-bit ADC(Analog to Digital converter) count, is converted to the voltage across a load resistor, then to MQ135 sensor resistance, normalized as R_s/R_0 , converted through an empirical power-law calibration to a gas-concentration proxy, and finally mapped to an estimated vitamin-C value. A separate first-order degradation model provides a time-based reference value. The two vitamin-C estimates are compared through a percentage-error calculation. The Temp sheet independently models cooling and humidity with exponential first-order responses.

2. Data Architecture and Why Each Stage Exists

Stage	Input	Equation / operation	Purpose
1. Measurement	Raw ADC count	ADC count is retained from the MQ135 gas sensor.	Direct digital representation of sensor voltage
2. Electrical conversion	ADC $\rightarrow$ voltage	$V = \text{ADC}/(\text{Max. ADC}) * V_{\text{ref}}$	Recover analog voltage represented by the ADC
3. Sensor resistance	Voltage $\rightarrow R_s$	$R_s = ((V_{\text{ref}} - V)/V)R_L$	Convert divider voltage into MQ135 resistance
4. Normalization	R_s and R_0	R_s/R_0	Remove absolute resistance scale for gas-sensor calibration
5. Ethylene Gas concentration (C_e , in ppm)	R_s/R_0	$C = a * (R_s/R_0)^b$	Empirical power-law calibration
6. Theoretical vitamin concentration (C_v , in ppm)	Time	$C_0 * \exp(-kt)$	First-order retention/degradation reference
7. Calculated vitamin concentration (c_v , in ppm)	C_v , C_e	$C_0 - m \ln(C_e + 1)$	Empirical mapping used by the prototype

8. Error	Two vitamin-C estimates	$ C_v - c_v /C_v \times 100$	Quantifies disagreement between the two model outputs
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3. Electrical Measurement Calculations

3.1 ADC count to voltage

Implemented equation:

$$V = (ADC / 1023) \times 5$$

Here

1023 = $2^{10}-1$, the sensor and microcontroller is a 10-bit ADC.

5 = V_{ref} , the voltage from the Arduino UNO board.

Example at $t = 0$ s: $ADC = 251 \rightarrow V = 251/1023 \times 5 = 1.226783969$.

3.2 Load-resistor / MQ135 resistance calculation

Implemented equation:

$$R_s = ((5 - V) / V) \times 10000$$

Here

5 = V_{ref} , the voltage from the Arduino UNO board.

10000 = load resistance with units Ω .

Example at $t = 0$ s, $R_s = ((5-1.226783969)/1.226783969) \times 10 = 30756.97211$.

3.3 Normalized sensor ratio

Implemented equation:

$$Ratio = R_s / R_o$$

Here

3.31 R0 Calibration

The $R_s(clean)=98.3$ k Ω and $R_s/R_0=3.6$. Therefore:

$$R_o = 98.3 / 3.6 = 27.305k\Omega$$

$$R_o = 27.305$$

Normalizing R_s by R_0 is conventional for metal-oxide gas-sensor calibration because the sensor response is commonly represented as a resistance ratio.

4. MQ135 Gas-Concentration Model

Implemented equation:

$$C_{VOC} = a \times (R_s/R_0)^b$$

4.1 Ethylene Power-Law Calibration

From the experimental data of two ethylene calibration points: 20 ppm at $R_s/R_0=0.65$ and 100 ppm at $R_s/R_0=0.35$.

So, $20 = a \times (0.65)^b$ and $100 = a \times (0.35)^b$

Therefore the equation is:

$$C_{ethylene} = 6.53 (R_s/R_0)^{-2.6} \text{ ppm}$$

This is a calibrated curve for the sensor. It is not a universal MQ135 equation. Power-law behavior is commonly used to represent metal-oxide gas-sensor response over an appropriate calibration range.

The negative power means that an increasing normalized resistance produces a decreasing calculated concentration. This is consistent with the general use of R_s/R_0 response curves for MQ-family sensors, but the numerical constants a and b are calibration parameters, not universal constants of the MQ135.

Example at $t = 0$ s: $R_s/R_0 = 30.75697/40 = 0.768924$.

$$\text{Therefore } C = 116.6(0.768924)^{-2.769} \approx 241.37 \text{ ppm}$$

5. First-Order Vitamin-C Retention Model

Implemented equation:

$$C_{ref}(t) = C_0 \times \exp[-k \times (t/86400)]$$

Here

$$C_0 = 103 \text{ ppm}$$

$$k = 0.045 \text{ day}^{-1}$$

t = time in seconds

This is the exponential solution of a first-order degradation model $dC/dt = -kC$. First-order kinetics are widely used to model ascorbic-acid/vitamin-C degradation under specified storage conditions.

Example at $t = 0$: $C_{ref} = 103.00$. At $t = 86400$ s (1 day): $C_{ref} = 103e^{-0.045} = 98.47$. The corresponding one-day retention is about 95.6% of the initial model value.

6. Derivation of the Gas-Sensor Vitamin-C Equation

In plant biology, degradation of organic compounds (like Vitamins) often follow an exponential (or) log decay pattern.

Since, the food can only decay, meaning the concentration of C_e strictly increases at the time increases, therefore we can relate concentration of vitamins with concentration of ethylene as,

$$C_v = 103 - m * \ln(C_e + 1)$$

The +1 prevents $\ln(0)$. This equation is not a fundamental chemical equation, but a calibrated gas-response proxy to the vitamin-C reference scale.

Here C_e = concentration of ethylene which changes with time t .

m = assumed to be constant, neglecting minor effects.

For each sample, the local coefficient is calculated as:

$$m = [103 - C_{ref,i}] / \ln(C_{ethylene} + 1)$$

From experimenting with different values of theoretical vitamin concentration and actual concentration, we get a value of m . The average of m is

0.3900579357.

The final equation :

$$C_v = 103 - 0.3900579357 \ln(C_e + 1)$$

7. Dataset Statistics

Metric	Result
Number of samples	86400
ADC	246 to 953; mean 599.486
V_RL	1.202 to 4.658 V
R_s	0.735 to 31.585 kΩ
Ethylene proxy	4.472 to 78982.248 ppm
C_ref	98.468 to 103.000
C_sensor	98.601 to 102.337
Signed mean error	0.0565 %
Mean absolute error	0.3131 %
RMSE of error	0.3550 %

The signed mean error is near zero because positive and negative errors cancel. For an accuracy statement, mean absolute error is the clearer metric: 0.313% for this dataset.

8. Thermal Model

8.1 Thermal Mass & Effective Conductance

The total thermal capacitance (C_{th}) is calculated by summing the heat capacities of the internal air volume, the primary aluminum heat sink, and the structural enclosure walls:

- Air Thermal Capacity: $C_{air} = \rho \times V \times C_v = 1.2 \text{ kg/m}^3 \times 0.03 \text{ m}^3 \times 718 \text{ J/(kg}\cdot\text{K)} = 25.85 \text{ J/K}$

- Heat Sink Capacity: $C_{sink} = m \times C_p = 0.15 \text{ kg} \times 900 \text{ J/(kg}\cdot\text{K)} = 135.00 \text{ J/K}$

- Wall Structural Capacity: $C_{wall} = 40.00 \text{ J/K}$

- Total Thermal Capacitance: $C_{th} = 25.85 + 135.00 + 40.00 = 200.85 \text{ J/K}$

The effective heat-transfer conductance (G) represents enclosure losses to ambient environment for a given temperature differential. With surface area $A = 0.59 \text{ m}^2$ and total thermal resistance $R_{total} = 0.2098 \text{ K/W}$:

$$G = A / R_{total} = 0.59 / 0.2098 = 2.8122 \text{ W/K}$$

8.2 TEC Steady-State Temperature

The steady-state energy balance for the thermoelectric cooler (TEC) incorporates enclosure losses, Joule heating, and conduction across the TEC elements:

$$T_{min} = [G \times T_{out} + (1/2) \times I^2 R + K \times T_h] / [G + \alpha I + K]$$

Given parameters: $T_{out} = 303.15 \text{ K}$, $T_h = 318.15 \text{ K}$, Seebeck coefficient $\alpha = 0.051 \text{ V/K}$, drive current $I = 4.2 \text{ A}$, internal resistance $R = 2.2 \text{ }\Omega$, thermal conductance $K = 0.6 \text{ W/K}$, and enclosure conductance $G = 2.8122 \text{ W/K}$.

Substituting these values yields:

$$T_{min} = [(2.8122)(303.15) + 0.5(4.2^2)(2.2) + (0.6)(318.15)] / [2.8122 + (0.051)(4.2) + 0.6] = 293.076 \text{ K (19.93 } ^\circ\text{C)}$$

8.3 Derivation of the Thermal Time Constant

For a lumped thermal system, the system time constant τ is the ratio of thermal capacitance to total effective thermal conductance (G_{eff}):

$$G_{eff} = G + \alpha I + K = 2.8122 + (0.051)(4.2) + 0.6 = 3.6264 \text{ W/K}$$

$$\tau = C_{th} / G_{eff} = 200.85 / 3.6264 = 55.3855 \text{ s}$$

9. Temperature Model

$$T(t) = T_{\infty} + (T_0 - T_{\infty})e^{-t/\tau}$$

Here

$$T_{\infty} = 293.076 \text{ K}$$

$$T_0 = 305.632 \text{ K}$$

$$\tau = 55.3855 \text{ s}$$

10. Scientific References

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